

Multiple Choice (10 marks)

1. Statement 1 | Every integral domain has a field of quotients. Statement 2 | A polynomial of degree n over a ring can have at most n zeros counting multiplicity.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1 | The homomorphic image of a cyclic group is cyclic. Statement 2 | The homomorphic image of an Abelian group is Abelian.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. For which value of n are there exactly two abelian groups of order n up to isomorphism?
 - (a) 4
 - (b) 7
 - (c) 8
 - (d) 12
4. Let $f(x, y) = x^3 + y^3 + 3xy$ for all real x and y . Then there exist distinct points P and Q such that f has a
 - (a) local maximum at P and at Q
 - (b) saddle point at P and at Q
 - (c) local maximum at P and a saddle point at Q
 - (d) local minimum at P and a saddle point at Q
5. Find the maximum possible order for an element of S_n for $n = 6$.
 - (a) 6
 - (b) 12
 - (c) 30
 - (d) 105

True / False (10 marks)

Answer True or False

6. True or False: The polynomial $4x^2 - 2$ is irreducible over the field of rational numbers $\mathbb{Q}$.
7. True or False: For any polynomials $f(x)$ and $g(x)$ in $\mathbb{R}[x]$, the degree of their product is equal to the maximum of their degrees.
8. True or False: The area of an equilateral triangle with an inscribed circle of radius 2 is 16.
9. True or False: There are infinitely many homomorphisms from $\mathbb{Z}$ into $\mathbb{Z}_2$.
10. True or False: The integers n that have only one group of order n (up to isomorphism) are 3, 5, 7, and 11.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the given system of linear equations and determine whether the statement "Every solution is a scalar multiple of $(-5, 1, 1, 0)$ " is true or false. Justify your answer with detailed reasoning.
12. Discuss the implications of two similar $n \times n$ invertible matrices A and B . Analyze how their similarity affects the matrices $A - 2I$, the trace, and the inverses A^{-1} and B^{-1} .

End of Paper